Engine Model(s): PT6A

72-50-04

POWER TURBINE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03 which is 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC06-004	Compound, Antiseize

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC30332	Wrench
PWC30335	Spreader
PWC30336	Sleeve
PWC30403	Puller
PWC30458	Squeezer

4. Fixtures, Equipment and Supplier Tools

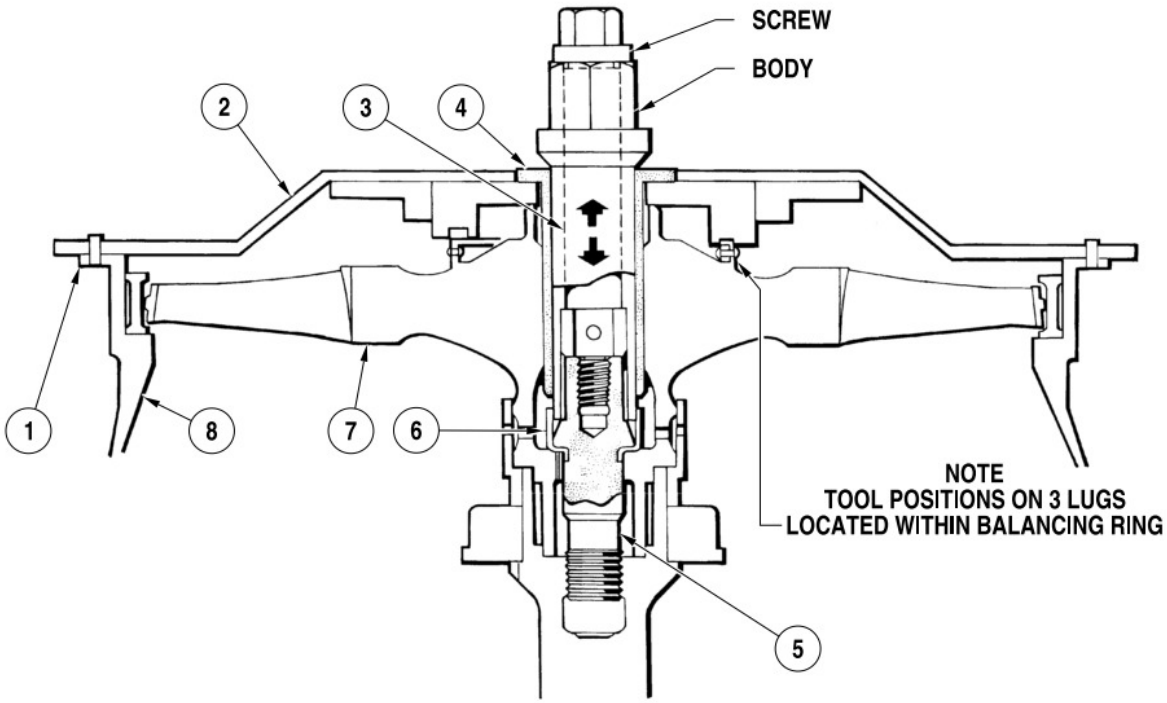
Not Applicable

5. Removal/Installation (Heavy Maintenance Only)

A. Removal of Power Turbine (Ref. Figs. 201, 202, 203 and 204)

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Figure 201 Unlocking Retaining Bolt Keywasher



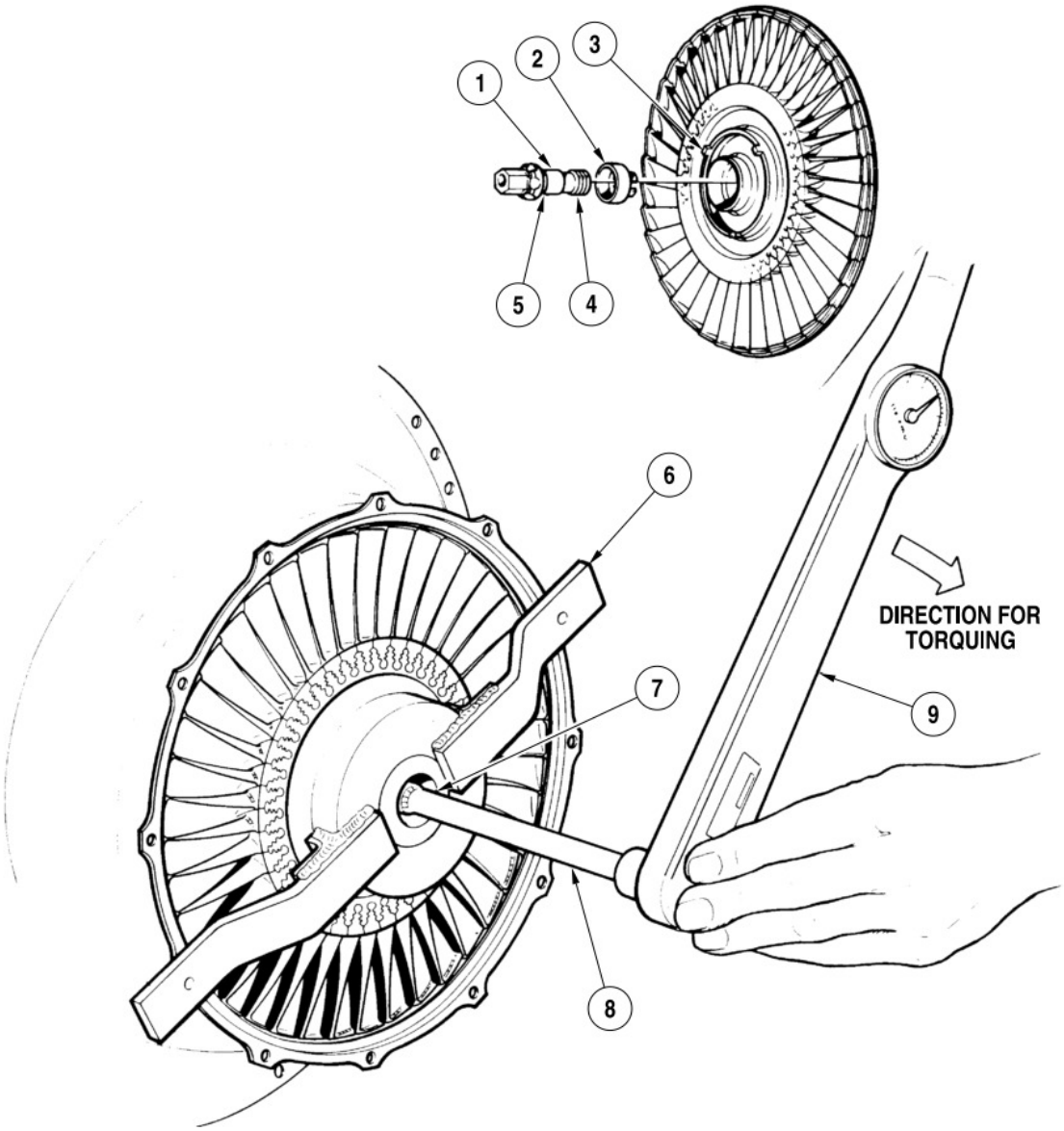
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1. Flange D
2. Wrench
3. Spreader
4. Protector Sleeve
5. Retaining Bolt
6. Keywasher
7. Power Turbine Disk Assembly
8. Exhaust Duct

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Figure 202 Removal/Installation of Power Turbine Retaining Bolt



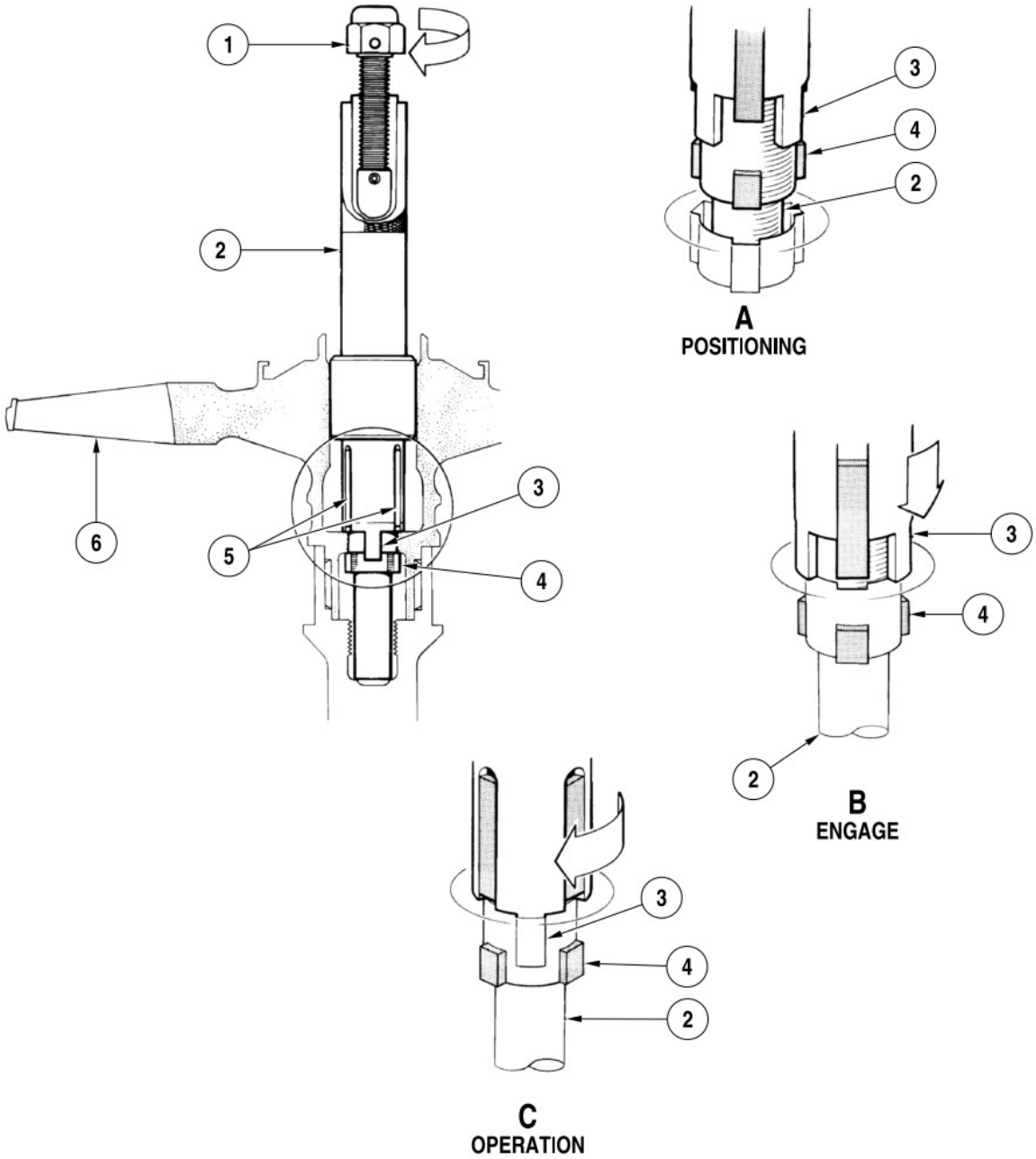
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1. Turbine Retaining Bolt
2. Keywasher
3. Lugs
4. Threads
5. Shoulder
6. Wrench
7. Protector Sleeve
8. Extension
9. Torque Wrench

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Figure 203 Removal of Power Turbine Disk



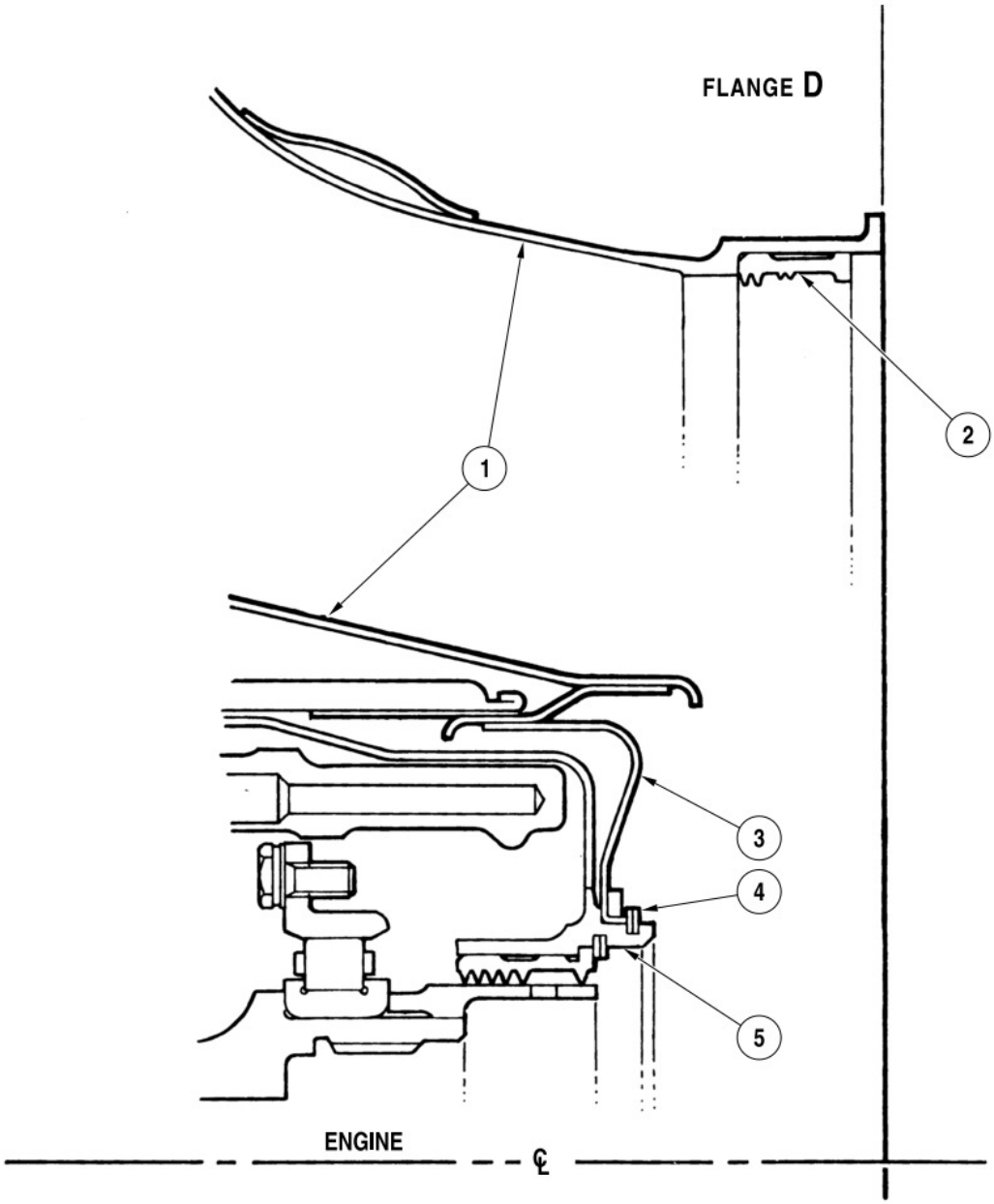
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1. Puller Screw
2. Puller
3. Lock
4. Legs
5. Slots
6. Turbine Disk

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Figure 204 Removal/Installation of No. 3 Bearing Cover



Pre-SB1430

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1. Exhaust Duct
2. Power Turbine Shroud
3. No. 3 Bearing Cover Assembly
4. Bearing Cover Retaining Ring
5. Power Turbine Shaft Housing Assembly

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(1) Remove the power section from the gas generator assembly and install in the stand (Ref. 72-50-03, Removal/Installation).

(2) Remove the power turbine vane ring and stator housing (Ref. 72-50-03).

(3) Install the wrench (PWC30332) (2, Fig. 201) on Flange D of the exhaust duct. Make sure the wrench slots are correctly seated over the lugs on the balancing ring of the disk (7).

(4) Insert the protector sleeve (PWC30336) (4) into the centerbore of the wrench.

(5) Turn the screw of the spreader (PWC30335) (3) until the shoulder seats on the body and insert the spreader into the protector sleeve to rest on the retaining bolt (5). Tighten the screw of the spreader to engage the bolt until resistance is felt.

(6) While holding the head of the screw with a conventional wrench, turn the body of the spreader with another conventional wrench to open the crimps on the keywasher (6). Remove the spreader.

(7) Using a conventional socket wrench, remove the retaining bolt (5) and keywasher (6). Remove the protector sleeve and wrench.

(8) Remove the power turbine disk using the puller (PWC30403) and conventional wrench (Ref. Fig. 203).

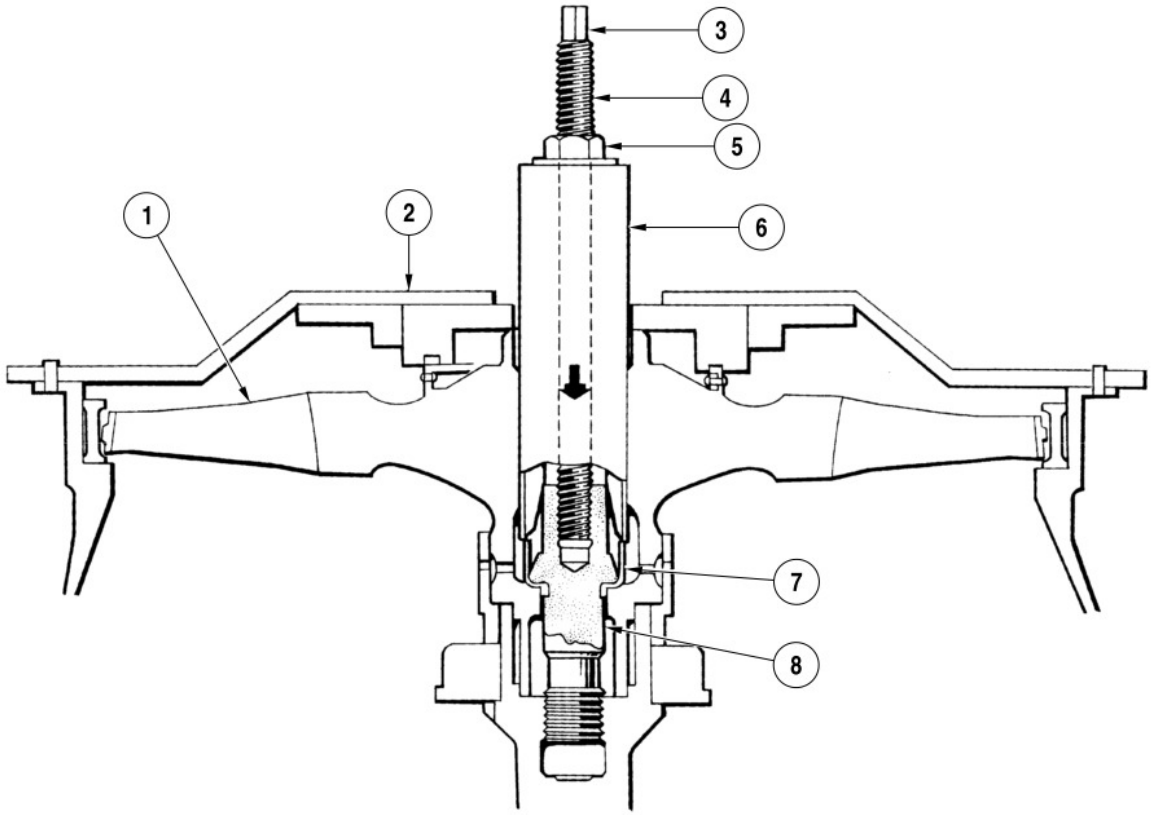
(9) If the No. 3 bearing cover assembly (3, Fig. 204) Pre-SB1430 shows evidence of wear or looseness, remove the retaining ring (4) securing the bearing cover assembly to the power turbine shaft housing assembly (5). Withdraw the No. 3 bearing cover assembly.

NOTE: On the exhaust duct assembly Post-SB1430, the No. 3 bearing cover is an integral part.

B. Installation of Power Turbine (Ref. Figs. 201, 202 and 205)

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Figure 205 Crimping Retaining Bolt Keywasher



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1. Turbine Disk
2. Wrench
3. Screw
4. Thread
5. Nut
6. Squeezer
7. Keywasher
8. Retaining Bolt

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(1) Install the No. 3 bearing cover assembly (3, Fig. 204) Pre-SB1430 over the end of the power turbine shaft housing assembly (5). Secure with the retaining ring (4).

NOTE: On the exhaust duct assembly Post-SB1430, the No. 3 bearing cover is an integral part.

(2) Apply a light film of the compound (PWC06-004) to the splines of the turbine disk, to the inside surface of the keywasher (2, Fig. 202) and to the shoulder (5) and threads (4) of the retaining bolt (1).

(3) Locate the turbine disk at flange D and align the master splines of the disk and power turbine shaft.

CAUTION: WHEN POSITIONING THE POWER TURBINE DISK, EXERCISE CARE NOT TO DAMAGE THE SHROUD LANDS BY CONTACT WITH THE TIPS OF THE DISK BLADES.

(4) Insert the keywasher (2, Fig. 202) and retaining bolt (1) into the centerbore of the disk. Make sure the key of the washer is correctly seated.

(5) Install the wrench (PWC30332) (6) and protector sleeve (PWC30336) (7) over the disk. Make sure the wrench slots are correctly seated over the lugs (3) of the disk and the dowels in the wrench are mated with the appropriate mounting holes on Flange D.

(6) Using a conventional socket, extension and torque wrench (Ref. Fig. 202), tighten the disk retaining bolt as follows:

(a) Initially torque 600 to 650 lb.in.

(b) Loosen to zero.

(c) For Pre-SB1459 Engines: Final torque 420 to 460 lb.in.

(d) For Post-SB1459 Engines: Final torque 500 to 550 lb.in.

(7) Remove the protector sleeve.

(8) Install the squeezer (PWC30458) (6, Fig. 205) in the centerbore of the wrench (2). Turn the screw (3) to engage the threads in the retaining bolt (8) until resistance is felt.

(9) Turn the nut (5) clockwise until the keywasher (7) is correctly crimped.

(10) Remove the squeezer (6) and wrench (2).

(11) Install the power turbine vane ring and stator housing (Ref. 72-50-03).

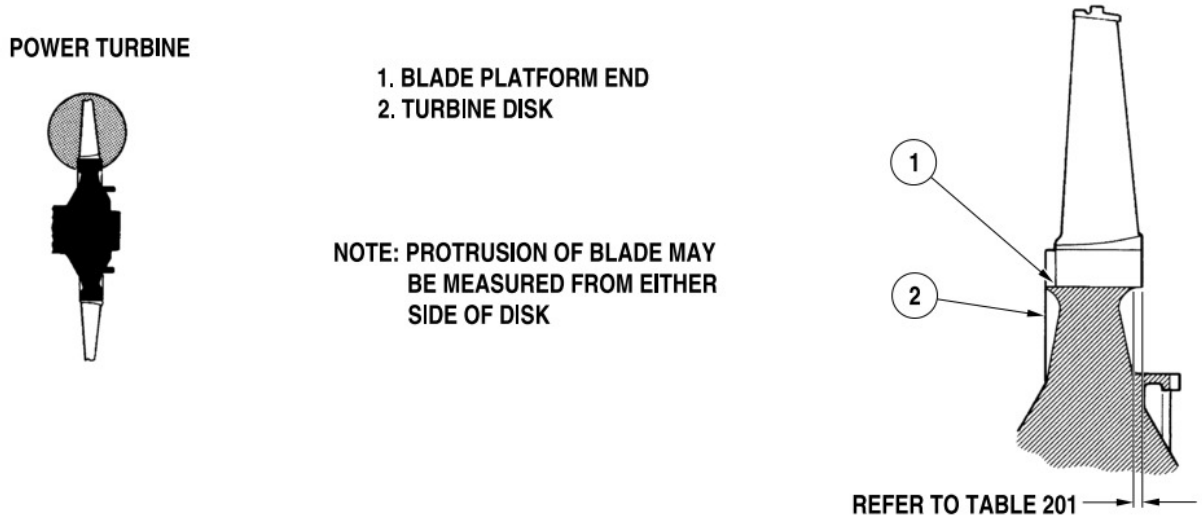
(12) Install the power section onto the gas generator (Ref. 72-00-00).

6. Inspection/Check (Heavy Maintenance Only)

A. Procedure (Ref. Figs. 206, 207 and Table 201)

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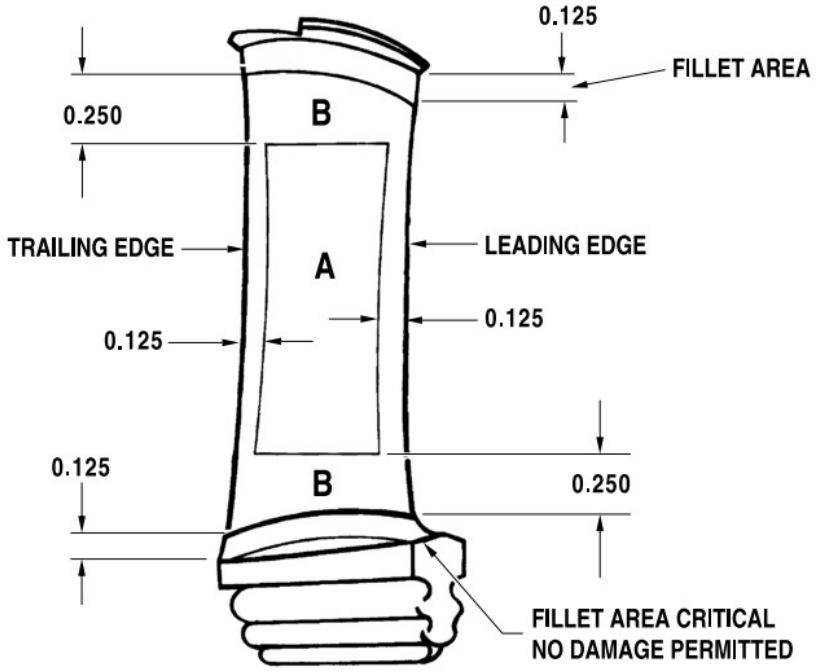
Figure 206 Power Turbine Blade Shift



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Figure 207 Power Turbine Blade Inspection



NOTE: RIPPLING OF TRAILING EDGE IS NOT ACCEPTABLE

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Table 201 Power Turbine Blade Inspection

Inspection	**Max. Serviceable**	**Action**
BLADE SHIFT	Protrusion of any blade root must be equal within 0.005 in. either side of disk (Ref. Fig. 206).	Replace *
BLADE LOOSENESS		
Circumferential movement	0.030 in. max. at tip	Replace *
Axial Movement	None permitted.	Replace *
AREA A:	(Ref. Fig. 207).	
Nicks	Three 0.003 in. deep max.	Replace *
Dents and pits	Three 0.003 in. deep max.	Replace *
Cracks	None permitted.	Replace *
AREA B:		
Nicks, dents and pits	One max., 0.020 in. deep max.	Replace *
LEADING AND TRAILING EDGES:		
Nicks, dents and pits	One max., 0.010 in. deep max.	Replace *
Cracks	None permitted.	Replace *
* Ship power section/turbine disks to an approved overhaul facility.		

(1) Examine the power turbine blades (Ref. Table 201).

(2) Inspect power turbine vane ring for condition and signs of damage, including nicks, dents, pits and minor cracks.

(3) Inspect turbine for condition and signs of damage. Inspection should be done by viewing through ports in exhaust duct.

(4) Inspect the No. 3 bearing cover assembly (Pre-SB1430). Replace the cover if evidence of rotation, rubbing or wear are noted.

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